

Product Requirements Document (PRD)

Project: Premium Interactive Greeting Card - "Cosmic Constellation V2" (Sequential Reveal)

1. Project Overview

We are building a premium, mobile-first interactive greeting card. This iteration (V2) pivots away from complex constellation drawing and focuses on a highly curated, cinematic, **Sequential Centerpiece Reveal**.

The recipient interacts with a starry night sky by tapping stars securely placed in the mobile "Thumb Zones". Each tap causes the star to explode and summons a memory into the exact center of the screen, guaranteeing 100% user focus.

2. The Recipient Experience (4-Phase Animation Sequence)

Phase 1: The Hook & Ignition

- **Visuals:** Deep space background with ambient Canvas starlight. A glowing nebula sits in the center. Elegant, glowing text at the top displays a dynamic hook_title (e.g., "✨ You are my Universe ✨").
- **Interaction (Press & Hold):** A glowing "Energy Core" sits in the lower center. The user must **press and hold** for ~1.5 seconds.
- **Feedback:** The core visually charges up (scales down, glowing ring appears).

Phase 2: The Star Map (Thumb-Zone Layout)

- **Action:** Upon ignition, the core and hook text fade out. The nebula clears.
- **Layout Constraints (CRITICAL):** 4 pulsing stars appear. To ensure flawless mobile accessibility, they must be hard-anchored to the left and right edges (The Thumb Zone):
 - Star 1: Left 15%, Top 28%
 - Star 2: Left 85%, Top 42%
 - Star 3: Left 20%, Top 68%
 - Star 4: Left 80%, Top 78%
- **Progress UI:** A sleek text prompt appears at the top: "Discover (0 / 4)".

Phase 3: The Sequential Memory Reveal (The Core Logic)

- **Action:** The recipient taps the stars in *any* order.
- **Effects on Click:**
 1. **Explosion:** The tapped star triggers an HTML5 Canvas particle burst and disappears.
 2. **The Modal:** A central glassmorphism modal fades in, dimming the background slightly (but keeping ambient stars visible).
 3. **The Payload:** The content shown is pulled from a **Smart Queue** (see section 3),

ensuring a cinematic build-up regardless of which physical star was tapped.

- **Memory UI Elements:**
 - **Photos:** Displayed in a beautifully framed vertical aspect ratio.
 - **Text:** Rendered using a "Silver Shimmering Foil" CSS effect. The text must appear using a JS Typewriter effect (approx. 80ms per character delay) while a CSS gradient shine continuously washes over it.
 - **Audio:** A premium pulsing circular waveform player.
- **Dismissal:** A pulsing "Tap anywhere to continue" hint appears at the bottom. Tapping closes the memory and returns them to the star map.

Phase 4: The Supernova Finale

- **Trigger:** After the 4th memory is closed, a glowing "Reveal Finale ✨" CTA button appears in the center.
- **The Blow Up:** Clicking the CTA triggers a massive "Supernova" flash (a pure white-out screen overlay).
- **The Final Card:** Under the cover of the flash, the Canvas background transitions from drifting stars to falling "Golden Cosmic Dust." As the white flash fades, a stunning, glowing holographic "Final Card" is revealed, displaying the Title Prefix, Recipient's Name, and Final Message.

3. The "Smart Queue" Payload Logic (CRITICAL DEVELOPER INSTRUCTIONS)

To maintain a cinematic story, the backend/frontend must build a sequential queue of exactly 4 items based on what the Sender uploaded. When a receiver taps a star, it pops the *next* item from this queue, not a hardcoded item tied to that specific star.

The Queue Assembly Rules:

1. Determine how many media items (Photos + Audio) the user uploaded.
2. Calculate Padding Needed = 4 - total_media.
3. Add plain text messages (fallback generic compliments) to the front of the queue to satisfy the padding.
4. Append any uploaded Photos (with their captions) to the queue.
5. Append the Audio note (if it exists) to the absolute end of the queue.

Expected Edge Cases (The 5 Scenarios):

- **Case 1 (0 Media):** User added nothing.
 - *Queue:* Text -> Text -> Text -> Text.
- **Case 2 (1 Photo):** User added 1 photo.
 - *Queue:* Text -> Text -> Text -> (Photo + Caption).
- **Case 3 (2 Photos):** User added 2 photos.
 - *Queue:* Text -> Text -> (Photo + Caption) -> (Photo + Caption).
- **Case 4 (3 Photos):** User added 3 photos.

- *Queue*: Text -> (Photo + Caption) -> (Photo + Caption) -> (Photo + Caption).
- **Case 5 (Max Capacity - 3 Photos + Audio):** * *Queue*: (Photo + Caption) -> (Photo + Caption) -> (Photo + Caption) -> Audio.

This guarantees that the emotional peak (the photos and the voice note) always happens at the very end of their interaction sequence, maximizing the emotional payoff.

4. Technical Stack Requirements

- **Framework:** React / Next.js.
- **Styling:** Tailwind CSS + Custom CSS for the Silver Foil Typewriter effect (background-clip: text, @keyframes shine).
- **Responsiveness:** Wrap the core experience in a max-w-[400px] container to enforce a mobile-screen aspect ratio even on desktop viewing, ensuring the 15% / 85% thumb-zone coordinates look proportional everywhere.
- **Canvas Engine:** Native HTML5 Canvas for:
 1. Ambient drifting stars (Phase 1 & 2)
 2. Tap explosions (Phase 3)
 3. Golden falling dust (Phase 4)

5. Tracking & Analytics

- Trigger memory_unlocked event for each star tap.
- Trigger finale_revealed event when the Supernova CTA is clicked.